

LendSignal

Updateable onchain credit certificates for B2B working-capital lending

Name recommendation

Use LendSignal for the hackathon: it is short, direct, and points to credit approval for business lending.

One-Liner

LendSignal turns a business wallet into an updateable onchain Credit Certificate, enabling working-capital loans through confidential verification, ENS-gated approval, and decentralized default protection.

Problem And Opportunity

- Emerging market businesses need working capital, but onchain lending still mostly requires excessive collateral.
- Protocols can see wallet activity, but not business documents, KYC/KYB evidence, repayment capacity, or fraud risk.
- Confidential AI can validate private borrower evidence and output only a score, risk tier, attestation hash, and evidence digest.
- A credit certificate lets lending vaults make repeatable credit decisions without exposing raw private data.

Product Flow

Step	What Happens	Public / Onchain Output
1. Onboard	Connect wallet, enter company data, submit mock docs and KYC/KYB.	Wallet becomes credit identity.
2. Verify	Confidential AI Attester-style flow analyzes borrower evidence.	AI score, risk tier, hashes.
3. Certify	Combine AI, bureau, and wallet-behavior signals.	Updateable Credit Certificate.
4. Lend	Vault checks certificate, score, ENS gate, and liquidity.	Automatic loan payout.
5. Protect	Borrower fees fund a default pool for LPs.	Lender reimbursement on default.

48-Hour MVP

Business onboarding	Wallet connect, business profile, mock document checklist, mock KYC/KYB status.
Credit score	Confidential AI score 70% + bureau score 20% + wallet behavior 10%.
Credit certificate	Registry stores score, risk tier, status, attestation hash, evidence digest, expiration.
Lending vault	Approves and pays out when certificate is active, score $\geq$ 750, ENS gate passed, and liquidity exists.
Default fund	LP deposits plus borrower fees; demo reimbursement on default.

Credit Certificate

The Credit Certificate is the public representation of the business credit state. It is updateable over time as new documents, bureau signals, wallet behavior, repayments, or defaults are observed.

Expose	business wallet, certificate id, combined score, risk tier, status, attestation hash, evidence digest, ENS gate, issued/expiry dates.
Hide	raw documents, bank statements, full KYC/KYB records, tax records, private invoices, personal identity documents.
Lifecycle	pending, active, updated, expired, revoked, defaulted.
Vault policy	approve if active, not expired, score $\geq$ 750, ENS gate passed, and vault has liquidity.

Chainlink Confidential AI Attester Integration

- Base URL: <https://confidential-ai-dev-preview.cldev.cloud>
- Flow: POST /v1/inference, then poll GET /v1/inference/:id until completed or failed.
- Resources: uploaded files or HTTP(S) URLs, up to 10 resources, each capped at 10 MiB.
- Models: gemma4 or qwen3.6.
- Hackathon fallback: if no key is available, use deterministic mock outputs with the same response shape.

Minimal Architecture

Component	Build For Hackathon	Why It Matters
Frontend	Onboarding, score, certificate, vault, default fund screens.	Makes the borrower and lender story obvious.
Score service	Mock or real attester API call, bureau mock, wallet behavior mock.	Creates the credit decision input.
Certificate registry	Issue, update, revoke, read certificate.	Turns credit state into reusable onchain primitive.
Lending vault	Deposit, request loan, approve and payout.	Demonstrates non-collateral-only lending.
Default fund	LP deposits, borrower fees, reimburse default.	Explains lender protection and business model.

Business Model

- Borrower certification fee: business pays to create or refresh a certificate.
- Loan origination fee: fee is charged when a vault pays out a loan.
- Default fund fee: part of borrower fees funds LP yield and lender protection.
- Protocol integration fee: lending pools pay for certificate checks, webhooks, or API access.
- Roadmap: agent-based monitoring, ERC-8004-style reputation, curator marketplace, real bureau integrations.

Pitch Close

LendSignal gives emerging market businesses a portable credit identity and gives lenders a safer way to approve undercollateralized working-capital loans.